

Revision · Sequences, the nth term and functions

Cambridge insert: the rule behind a pattern, written once and used for every term.

LESSON 91–93

3 × 40 MIN

ALGEBRA 8 – REVISION

STAGE 9 · 9.1–9.3

LESSON CLOCK

6'

12'

10'

8'

4'

Continue three patterns	6 min
The nth term	12 min
Quadratic and special sequences	10 min
Function notation	8 min
Homework	4 min

LEARNING OBJECTIVES

- Continue a sequence and describe its term-to-term rule.
- Find the nth term of a linear sequence.
- Recognise square, cube and simple quadratic sequences.
- Use function notation and complete a table of values.

TEXTBOOK

National	Annual revision
Cambridge	Learner’s Book pp. 194–210
Workbook	Workbook 9.1–9.3

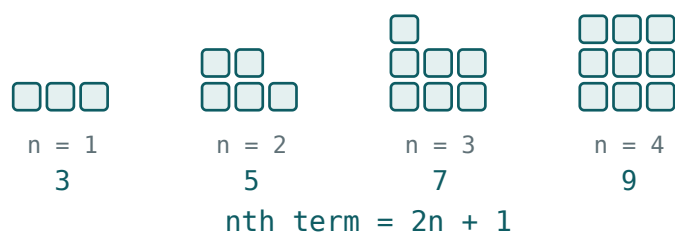
01

Explanation

Two ways to describe a sequence

Look at 5, 8, 11, 14, 17, ...

RULE	SAYS	GOOD FOR
term-to-term	“add 3 to the term before”	the next term
position-to-term (nth term)	$3n + 2$	the 100th term, straight away



A linear sequence grows by the same amount each step. The common difference is the coefficient of n in the n th term.

NTH TERM OF A LINEAR SEQUENCE

$n\text{th term} = dn + (a_1 - d)$

where d is the common difference and a_1 the first term. In practice: write $d \cdot n$, then adjust by whatever is needed to make $n = 1$ come out right.

For 5, 8, 11, ... the difference is 3, so start with $3n$: that gives 3, 6, 9, ... — every term is 2 too small. The rule is $3n + 2$. Check at $n = 4$: $12 + 2 = 14$ ✓.

Sequences that are not linear

SEQUENCE	PATTERN	NTH TERM
1, 4, 9, 16, 25	squares	n^2
1, 8, 27, 64	cubes	n^3
2, 4, 8, 16, 32	doubling	2^n
3, 6, 11, 18, 27	squares shifted	$n^2 + 2$
1, 3, 6, 10, 15	triangular	$\frac{1}{2}n(n + 1)$

The test is the **second difference**. If the differences themselves change by a constant, the sequence is quadratic:

$3, 6, 11, 18, 27 \rightarrow$ first differences $3, 5, 7, 9 \rightarrow$ second differences $2, 2, 2$

A constant second difference of 2 means an n^2 sits at the heart of the rule.

Function notation

A **function** is a machine: one input goes in, exactly one output comes out. Writing $f(x) = 3x + 2$ says the machine multiplies by 3 and then adds 2.

$f(0) = 2, f(4) = 14, f(-1) = -1$

A table of values collects several inputs at once, which is exactly what you need before drawing a graph:

x	-2	-1	0	1	2	3
$3x + 2$	-4	-1	2	5	8	11

SEQUENCE OR FUNCTION?

A sequence takes only whole-number positions $n = 1, 2, 3, \dots$ — its graph is a row of dots. A function like $f(x) = 3x + 2$ accepts any x , so its graph is an unbroken line.

02

Worked examples

EXAMPLE 1 Find the n th term of 7, 12, 17, 22, ... and use it to find the 30th term.

1

$d = 5$
Each term is 5 more than the one before.

2

$5n$ gives 5, 10, 15, ...
Every term is 2 too small.

3

n th term $= 5n + 2$
Check $n = 3$: $15 + 2 = 17$ ✓

4

$a_{30} = 150 + 2 = 152$

ANSWER

$5n + 2$, 30th term 152

EXAMPLE 2 Find the n th term of 4, 7, 12, 19, 28, ...

① *first differences* 3, 5, 7, 9
Not constant, so not linear.

② *second differences* 2, 2, 2
Constant — quadratic, with n^2 inside.

③ n^2 gives 1, 4, 9, 16, 25
Compare term by term.

④ *difference is 3 every time*

⑤ n th term = $n^2 + 3$
Check $n = 5$: $25 + 3 = 28$ ✓

ANSWER

$$n^2 + 3$$

EXAMPLE 3 If $f(x) = 2x - 5$, find $f(3)$, $f(-2)$ and the x for which $f(x) = 11$.

① $f(3) = 6 - 5 = 1$
Substitute.

② $f(-2) = -4 - 5 = -9$

③ $2x - 5 = 11$
Now run the machine backwards.

④ $2x = 16$, $x = 8$

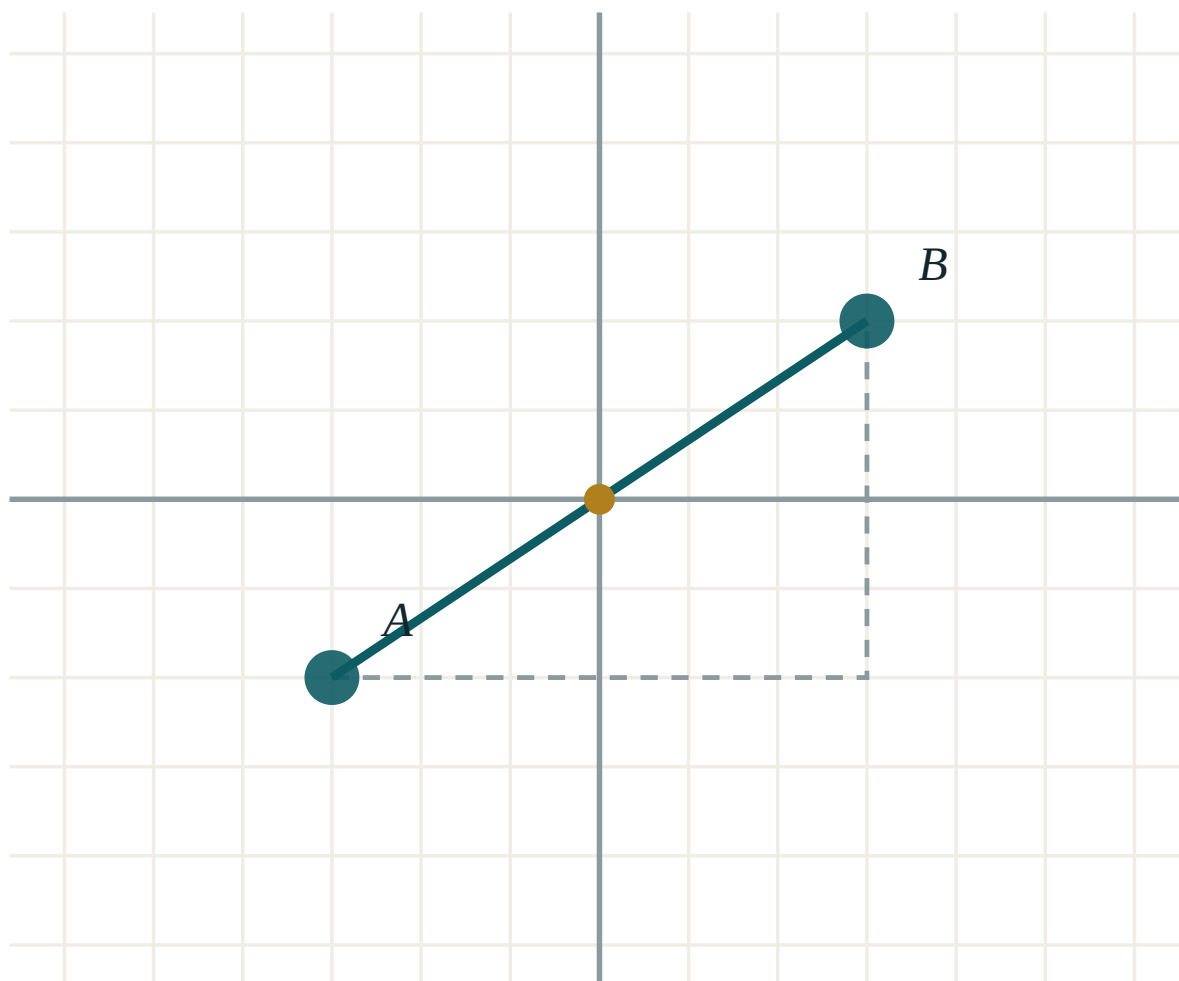
ANSWER

$$f(3) = 1, f(-2) = -9, x = 8$$

03 Interactive model

Drag the points to see how the common difference controls the steepness of a linear pattern.

A sequence plotted as points



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sequence	Ketma-ketlik	Последовательность
Term	Had	Член (последовательности)
Term-to-term rule	Hadlar orasidagi qoida	Рекуррентное правило
Position-to-term rule	O'rin bo'yicha qoida	Формула n-го члена
nth term	n-had	n-й член
Common difference	Ayirma	Разность
Linear sequence	Chiziqli ketma-ketlik	Линейная последовательность
Function	Funksiya	Функция
Input / output	Kirish / chiqish	Вход / выход
Table of values	Qiyimatlar jadvali	Таблица значений

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The n th term of 4, 9, 14, 19, ... is:

The 20th term of $3n - 2$ is:

A constant second difference means the sequence is:

If $f(x) = 5 - 2x$ then $f(-3)$ is:

-1

11

-11

1

Which sequence has n th term $n^2 + 1$?

$1, 4, 9, 16$

$2, 5, 10, 17$

$2, 4, 6, 8$

$1, 3, 5, 7$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Write the next two terms of 3, 7, 11, 15, ...

ANSWER 19, 23

2. Find the n th term of 2, 4, 6, 8, ...

ANSWER $2n$

3. Find the n th term of 5, 10, 15, 20, ...

ANSWER $5n$

4. Find the 10th term of $4n - 1$.

ANSWER 39

5. If $f(x) = x + 7$, find $f(3)$.

ANSWER 10

6. If $f(x) = 3x$, find $f(-2)$.

ANSWER -6

7. Write the first four terms of n^2 .

ANSWER 1, 4, 9, 16

Medium · the standard question

7 PROBLEMS

1. Find the n th term of 6, 11, 16, 21, ...

ANSWER $5n + 1$

2. Find the n th term of 20, 17, 14, 11, ...

ANSWER $23 - 3n$

3. Find the 50th term of $7n - 3$.

ANSWER 347

4. Find the n th term of 2, 5, 10, 17, 26, ...

ANSWER $n^2 + 1$

5. If $f(x) = 4x - 3$, solve $f(x) = 17$.

ANSWER $x = 5$

6. Make a table of $2x + 1$ for $x = -2, -1, 0, 1, 2$.

ANSWER $-3, -1, 1, 3, 5$

7. Is 61 a term of $4n + 1$? Justify.

ANSWER Yes — $4n = 60$, $n = 15$, a whole number.

Hard · stretch and reason

7 PROBLEMS

1. Find the n th term of 3, 8, 15, 24, 35, ...

ANSWER $n^2 + 2n$ (i.e. $n(n + 2)$)

2. Is 100 a term of $3n + 2$? Justify.

ANSWER No — $3n = 98$ is not a multiple of 3.

3. The 4th term of a linear sequence is 17 and the 9th is 37. Find the n th term.

ANSWER $d = 4$, n th term $= 4n + 1$

4. Find the n th term of 1, 3, 6, 10, 15, ...

ANSWER $\frac{n(n + 1)}{2}$

5. If $f(x) = x^2 - 4x$, find all x with $f(x) = 0$.

ANSWER $x = 0$ or $x = 4$

6. A sequence starts 2, 6, 12, 20, 30. Find the 20th term.

ANSWER $n(n + 1)$, so 420

7. Explain why a linear sequence can never contain both 7 and 8 if its common difference is 3.

ANSWER Consecutive terms differ by 3, so no two terms differ by 1; every term leaves the same remainder on division by 3, and 7 and 8 do not.

07 Homework

Homework — 6 problems

Cambridge Stage 9, Unit 9. Always check your n th term at $n = 1$ and $n = 3$.

- Find the n th term of 8, 13, 18, 23, ...
- Find the 40th term of $6n - 5$.
- Find the n th term of 3, 6, 11, 18, 27, ...
- If $f(x) = 5x - 4$, find $f(0)$, $f(3)$ and the x with $f(x) = 26$.

5. *Make a table of $3 - 2x$ for $x = -2 \dots 2$.*

6. *Is 73 a term of $5n + 3$? Explain.*